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# Can This Agent Even Do That?

## A Decidable Goal-Achievability Type Discipline for LLM-Synthesized Agent Skills

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### Abstract

1 LLM agents are increasingly driven by natural-language specifications, expressed  
2 through artifacts such as `SKILL.md` and `AGENT.md` files that describe an agent’s  
3 capabilities, tools, and workflows. Agent Skills, for example, package instructions  
4 that guide an agent in performing specific tasks [5]. But how can we know whether  
5 such specifications are themselves correct? Even if an LLM agent follows a speci-  
6 fication exactly, can it actually achieve the goal that the specification is intended to  
7 accomplish? Or might the goal be fundamentally unachievable, discovered only  
8 after spending substantial time and tokens executing the agent—or even ending  
9 in deadlock? We present a *skill compiler* that formally validates, before runtime  
10 execution and without relying on an LLM judge, whether a skill’s *goal is achiev-*  
11 *able*. The compiler combines capability-guarded reachability from automated  
12 planning with deadlock-freedom from **simple multiparty sessions (SMPS)**, decided  
13 *directly* over session configurations and a goal-marked global type synthesized  
14 from the natural-language specification. Our approach follows the **SMPS** discipline  
15 of checking a session directly against a global protocol [1, 2]. The approach begins  
16 by using an LLM to extract the content of a natural-language specification and  
17 compile it into a formal logical pack consisting of preconditions, effects, and goals,  
18 which are then consumed by the inference rules. The checker is specified by a  
19 Coq-verified, refutation-sound abstraction theorem. We further establish subject  
20 reduction and session fidelity for the direct discipline. The single-label commu-  
21 nication fragment with bystander interleaving is mechanized, while world-changing  
22 action interleaving is proved under explicit hypotheses of goal stability and effect  
23 commutativity. We formally define the language and its behavior, specify how  
24 skills and sessions are checked, and prove that the resulting checking framework  
25 has several important correctness properties. We also identify which part of the  
26 system remains decidable and show where the boundary of undecidability begins  
27 when participants can be added dynamically.

## 28 1 Introduction

29 A modern agent is no longer merely a chatbot, but an autonomous system capable of taking actions  
30 beyond text generation—for example, writing code, sending emails, or executing trades. Such  
31 behaviour is often guided by natural-language artifacts that specify instructions, workflows, and  
32 constraints for the agent [3, 4, 6, 7]. A *skill* is one common way of expressing such guidance. It is a  
33 folder containing instructions and tooling that an agent can discover and load as needed [5], thereby  
34 *harnessing* a general-purpose agent toward a particular outcome.

35 There is growing evidence that this guidance matters. Across seven state-of-the-art open-source  
 36 multi-agent systems, failure rates range from 41% to 86.7% [8]. More importantly, some of these  
 37 failures can be addressed without changing the underlying model. In one system, simply adding a  
 38 task-objective verification step to the specification improves task success by 15.6% [8, Appendix H].  
 39 Likewise, repairing the harness based on execution traces improves task completion by 6.3 to 18.4  
 40 percentage points across four agent benchmarks [4]. These results show that the instructions and  
 41 structure surrounding an agent can matter as much as the model itself.

42 The question, however, is how we can know whether such guidance is sufficient to achieve the  
 43 outcome it declares. A plan may *book a flight and email a confirmation* even though no email tool is  
 44 available. It may pursue *a flight under \$500* even though the available booking tool can return only  
 45 premium fares. Or a planner may wait for a worker’s result while the worker waits for the planner’s  
 46 answer, with each expecting a message that the other will never send. These are not merely execution  
 47 failures. In each case, the goal is already unreachable before execution begins.

48 Such failures are structural and recurring, yet today the only way to discover them is often to run  
 49 the skill and inspect what went wrong afterwards [4]. By then, the agent may already have spent  
 50 substantial time and tokens taking actions, and a multi-agent system may already have become stuck  
 51 waiting for communication that will never occur. A simple static check could, in principle, identify  
 52 these problems before execution.

53 This paper asks a deliberately narrow question and answers it with a deterministic formal typing  
 54 system: **given an LLM skill, can its goal be achieved at all?** We do not, here, verify that every step  
 55 is safe, nor that the agent behaves correctly on every possible input. Instead, we decide *achievability*.  
 56 The analysis is tolerant: small details, detours, and payload variation should not trigger a false alarm.  
 57 At the same time, it is *sound for refutation*: when the compiler concludes that a goal is impossible,  
 58 that verdict is correct.

## 59 1.1 The idea, and the trust boundary

60 The architecture, shown in Figure 1, has two parts separated by a trust boundary. First, an LLM  
 61 performs *compaction*: it reads the natural-language skill and turns it into a formal *pack* containing  
 62 capabilities with preconditions and effects, a goal-marked global protocol, and an initial state. This  
 63 step is *untrusted*: the LLM may misunderstand the skill or omit relevant details.

64 A deterministic *checker* then decides achievability using only the resulting pack. The checker is  
 65 mechanically proved sound: if it returns *impossible*, then the goal is impossible under the formal pack  
 66 it received. If the LLM omits a constraint or capability from the original skill, the checker cannot  
 67 recover information that is not present in the pack; it will instead reason about the incomplete pack.  
 68 Conversely, if the LLM introduces a constraint that was not in the original skill, the checker may  
 69 report the resulting goal as impossible. Thus, our guarantee is about the checker’s verdict with respect  
 70 to the formal pack, not about whether the pack perfectly represents the original natural-language skill.

71 This does not make the compaction step a weakness that we simply ignore. The formal pack provides  
 72 a structured target for the LLM to extract. Rather than asking the LLM to freely invent a formal  
 73 model, we give it a predefined logical structure: capabilities have preconditions and effects, goals are  
 74 explicit, and the protocol follows a fixed syntax. This makes compaction a constrained translation  
 75 task and gives us a concrete artifact that can be inspected before execution.

76 Crucially, the gap between *what the user meant* and *what was formalized* cannot be eliminated. We  
 77 instead make this gap explicit at the compaction step, where the resulting pack can be inspected and  
 78 checked before execution. When the compaction is accurate, the checker can identify goals that are  
 79 impossible before the agent spends time and tokens executing the skill. When the compaction is  
 80 inaccurate, the checker still provides a sound verdict for the formal pack it received. The key design  
 81 idea is therefore to move achievability checking from runtime execution to an explicit, inspectable  
 82 checkpoint before execution.

## 83 1.2 Contributions

- 84 1. **A goal-achievability type discipline** (Section 3 and Appendices A–C) that combines  
 85 capability-guarded reachability from automated planning with deadlock-freedom from  
 86 multiparty session types. The discipline checks whole session configurations directly against

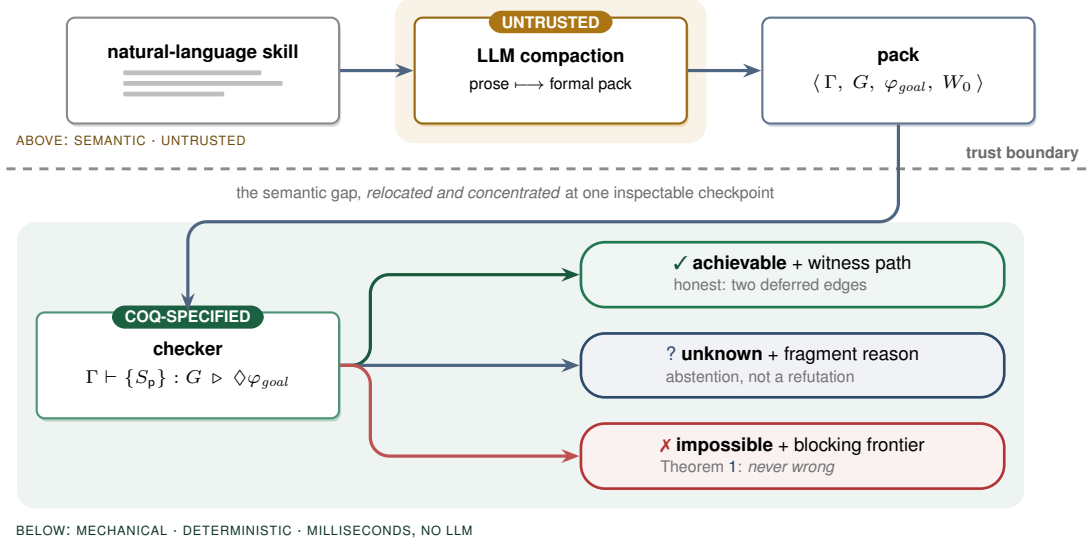


Figure 1: The trust boundary. An untrusted LLM compacts natural-language instructions into a formal pack. A deterministic checker then decides achievability over the pack using a Coq-verified abstraction theorem. The verdicts are asymmetric: *impossible* is sound (Theorem 1), while *achievable* depends on the accuracy of the pack (Section 3.1).

- 87 a goal-marked global type synthesized from the skill. Building on the multiparty session  
88 type tradition of Honda, Yoshida, and Carbone [12, 13], and the direct session-checking  
89 approach of **SMPS** [1, 2], it adds capabilities, explicit goals, and a refutation-sound but  
90 achievement-incomplete analysis for LLM-drafted skills.
- 91 2. **A mechanized soundness specification** (Appendix D) formalized in Coq. We prove a  
92 refutation-sound abstraction theorem parameterized by simulation hypotheses, together  
93 with tolerance soundness and capability monotonicity: adding capabilities cannot make an  
94 achievable goal impossible. We also establish operational correspondence through subject  
95 reduction and session fidelity for the direct judgment over a labelled transition system. For  
96 single-label communication, we mechanize the cases where other participants continue to  
97 act concurrently; the world-changing action cases are proved on paper under explicit side  
98 conditions.
  - 99 3. **A decidability boundary** (Appendix D). Achievability is decidable when the set of par-  
100 ticipants is fixed. We also show that an extension allowing the dynamic addition of an  
101 unbounded number of participants becomes undecidable.
  - 102 4. **An implementation and evaluation** (Sections 4–5). We implement the checker together  
103 with an LLM-compaction front-end and a schema gate, and evaluate it on a corpus covering  
104 the main failure modes. The results are consistent with the theoretical guarantees: we  
105 observe no false impossibility verdicts, while every false achievability verdict falls into one  
106 of the two deferred cases.

## 107 2 Overview by Example

108 Consider a one-agent skill whose goal is *a flight is booked and a confirmation is sent*. The LLM  
109 compacts the skill into a formal pack. If no email tool is available, there is no action that can establish  
110 confirmation\_sent.

$$\begin{aligned}
 \varphi_{goal} &= \text{booked} \wedge \text{confirmation\_sent} \\
 \Gamma(a) &= \langle pre_a, \langle Add_a, Del_a, Asg_a, Nd_a \rangle \rangle \\
 \text{search} &\mapsto \langle \top, \langle \{\text{searched}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\
 \text{filter} &\mapsto \langle \text{searched}, \langle \{\text{filtered}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\
 \text{book} &\mapsto \langle \text{filtered}, \langle \{\text{booked}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\
 \text{email} &\mapsto \langle \text{booked}, \langle \{\text{confirmation\_sent}\}, \emptyset, \emptyset, \emptyset \rangle \rangle
 \end{aligned}$$

$$\text{dom}(\Gamma_A) = \{\text{search}, \text{filter}, \text{book}\} \quad \text{dom}(\Gamma_B) = \text{dom}(\Gamma_A) \cup \{\text{email}\}$$

Under STRIPS [17] frame semantics, the checker therefore returns *impossible* and identifies the missing capability. If the email tool is available, the goal is reachable through the sequence  $\text{search} \cdot \text{filter} \cdot \text{book} \cdot \text{email}$ , which the checker returns as a witness. The same example is mechanized in Appendix D (the instance `FlightInstance`). Importantly, the booking itself remains reachable in the first case; the refutation concerns only the missing capability needed to complete the goal.

Tolerance is also important. A skill should not be rejected simply because an agent sends extra status messages, takes a clarifying step, or uses a different payload. Our analysis therefore allows these variations in three ways: *existential may-reachability*, where one successful path is sufficient; *session subtyping*, which allows safe interface variation; and *goal-relevant abstraction*, which ignores payload details that do not affect the goal. These mechanisms are defined in Appendices A–C.

### 3 Verification Model and Signals

**Declared pack.** The checker consumes  $P = \langle \Gamma, G, \varphi_{\text{goal}}, W_0 \rangle$ . Here,  $\Gamma$  contains capability declarations with guards and STRIPS-style effects;  $G$  is a goal-marked global protocol (a *global type*; Appendix A);  $\varphi_{\text{goal}}$  is the logical goal; and  $W_0$  is the initial abstract world state. The LLM produces  $P$  from the natural-language skill. A deterministic schema gate enforces the decidable restrictions: finite static roles, guarded recursion, finite-domain booleans, and quantifier-free linear integer arithmetic. The formal syntax and semantics are given in Appendices A–B.

**Verification signals.** The checker uses three static signals:

| Signal                        | Role                        | Output                                   |
|-------------------------------|-----------------------------|------------------------------------------|
| tool availability and effects | capability reachability     | missing capability or witness action     |
| coordination structure        | global protocol conformance | protocol mismatch or conformant protocol |
| goal and cost refinements     | logical reachability        | blocked guard or model                   |

These signals are checked before execution. Runtime payload checks remain separate obligations and are not assumed by the static checker.

**Decision procedure.** After the schema gate, the checker (i) rejects undeclared capabilities and goal conditions that cannot be achieved by any available capability; (ii) checks role behaviors against  $G$ , requiring exact sender choices while allowing safe receiver-side branch supersets; and (iii) explores existential protocol/world successors from  $W_0$ , using Z3 [18] for guards and refinements. A goal condition that cannot be satisfied blocks the search. If the solver times out or the pack falls outside these restrictions, the checker returns UNKNOWN rather than *impossible*. The checker therefore returns:

|                        |                                                        |
|------------------------|--------------------------------------------------------|
| IMPOSSIBLE( $F$ )      | a sound explanation of why the goal cannot be reached, |
| ACHIEVABLE( $\pi, O$ ) | a witness path $\pi$ and deferred obligations $O$ ,    |
| UNKNOWN( $r$ )         | an abstention with a reason $r$ .                      |

Each verdict records the pack digest, checker semantics, and artifact version.

**Guarantees.** Let a configuration consist of a protocol state and an abstract world state, and let *abs* map concrete configurations to abstract configurations. Assume step and goal simulation for the declared pack (Appendix D). If no abstract run reaches  $\varphi_{\text{goal}}$ , no concrete run represented by that pack reaches the concrete goal (Theorem 1). The analysis remains sound when the abstraction is made less detailed, and adding capabilities cannot turn an achievable goal into an *impossible* one (Theorems 2–3). Direct conformance satisfies subject reduction and session fidelity (Theorems 4–5). The Coq development covers single-label communication with concurrent actions by other participants; world-changing interleavings require explicit stability and commutativity assumptions. The decidable restrictions above guarantee decidability. Allowing an unbounded dynamic addition of participants is an explicit undecidable extension, for which the implementation returns UNKNOWN. Full rules, assumptions, proofs, and mechanization scope appear in Appendix D.

### 3.1 Structured feedback and human oversight

The checker returns an explanation for IMPOSSIBLE and a witness path for ACHIEVABLE. These outputs provide concrete feedback rather than a single score. A bounded repair adapter may ask the LLM to fix a protocol, but it cannot invent capabilities or weaken the goal. Every revision is checked again by the deterministic checker. Human review remains necessary for the gap between the original intent and the formal pack. Automated prompt or scaffold search is an application of these verdicts, not an empirical contribution of this paper.

## 4 Implementation

We implemented the checker and its supporting components in Python. The compaction front-end sends the skill to an LLM and passes the result through a deterministic *schema gate* before the checker processes it. Packs that do not satisfy the schema are rejected at this stage. The checker implements the abstractions from Appendices A–C: STRIPS frame semantics for the world, existential search over the protocol, and Z3 for guard and goal satisfiability and arithmetic refinements. On success it returns a witness path; on refutation, it reports the reason, such as a missing capability, unsatisfiable guard, unachievable goal condition, or non-projectable handoff. The optional LLM front-end may use a NON\_PROJECTABLE counterexample for one bounded repair round. It may not invent capabilities or weaken the goal, and the deterministic checker remains the final decision maker.

The conformance step ( $G \vdash (\mathbb{M}; W)$ , Section C.2) is **implemented by means of a simple type inference algorithm since all typing rules are structural in  $(\mathbb{M}; W)$ . At each step we freely choose either one or two participant processes to reduce or a goal satisfied by the world obtaining the set of sessions which must be typed as premises of the typing rule.**

The adapter also checks the participant side conditions of T-COMM/T-ACT/T-GOAL: it computes  $\text{prt}(G)$  and compares it with the roles for which the pack declares behaviour. A declared role outside  $\text{prt}(G)$  has contract End, so a non-trivial behavior for that role cannot conform. If the pack declares no behavior for a participant, the checker reports the missing role instead of leaving the assumption implicit.

## 5 Evaluation

We assembled a corpus of 15 skills covering the main failure modes: hallucinated planning, retry-forever loops, coordination deadlock, and refinement violations. The corpus also includes achievable controls with detours and informed branches, plus two deliberately *spurious* cases for the deferred checks. Each skill has a ground-truth semantic label; the positive class is *achievable*. Evaluation is deterministic and CPU-only: it requires no training or live model inference, completes the 15-case matrix in under one second on a commodity laptop, and gives each individual Z3 query a 10-second timeout.

|                  | predicted achievable | predicted impossible |
|------------------|----------------------|----------------------|
| truly achievable | TP = 6               | FN = 0               |
| truly impossible | FP = 2               | TN = 7               |

Two claims, both confirmed. **Soundness:** FN = 0 — no achievable goal was wrongly refuted, consistent with Theorem 1 (achievability recall  $6/6 = 100\%$ ). **Incompleteness, contained:** FP = 2, and both are the planted spurious cases — one a false payload post-condition (Layer-C obligation), and one an under-specified goal (intent edge). No structural failure was missed in this proof-of-concept corpus. This is evidence of coverage, not a proof that these are the only possible failure modes. Each refutation reason matches its intended failure mode:

| Failure mode (source)                                      | Verdict reason     |
|------------------------------------------------------------|--------------------|
| hallucinated planning — invokes a non-existent tool        | MISSING_CAPABILITY |
| hallucinated planning — no establisher for a goal conjunct | GOAL_UNSAT         |
| retry-forever loop — guard never satisfiable               | BLOCKED_GUARD      |
| coordination deadlock — unobserved choice / bad handoff    | NON_PROJECTABLE    |
| refinement violation — budget unsatisfiable on every run   | GOAL_UNSAT         |

**What the check costs.** The checker and deterministic front-end use *zero* LLM tokens. The LLM is used once to compact each skill *version*, while a full agent run uses tokens on every *invocation*. Under the conservative model in Appendix E, compacting the 15-spec corpus costs 12.2K tokens and avoids  $\sim 1.07\text{M}$  tokens per round of invocations, giving  $\sim 87\times$  leverage. For every failure mode, the cost of compaction is recovered by the first prevented run. For a skill that is *not* broken, the check costs 2.9% of one successful run, or nothing.

A separate six-case suite tests behavior not covered by the main matrix: tail-recursive retry and non-terminating control, adding new participants during execution, refutation that remains valid when new participants are added, and conformant versus non-conformant declared role behavior. It yields two ACHIEVABLE, three IMPOSSIBLE, and one UNKNOWN. The abstention is reported separately rather than counted as a false negative because UNKNOWN makes no refutation claim. Together the suites contain 21 declared packs; the 15-case matrix remains the only binary ground-truth matrix.

## 6 Related Work

*Multiparty session types* [13, 16, 9] provide formal methods for checking communication protocols, including deadlock-freedom through projection and subtyping [14, 10]. Our protocol follows the synchronous formulation of this line of work [9–11]. We also follow the recent approach of checking a whole session configuration directly against a global type [1, 2], rather than first projecting the global type to local types. The new contribution is the world and goal layer: capability guards, effects, goal observation, and the asymmetric refutation guarantee are not modelled by classical MPST or new SMPS. These are not part of classical MPST [13, 16, 9] or the recent direct-checking formulation [1, 2].

*Automated planning* [17] provides methods for deciding whether a goal can be reached from an initial state using available actions. We use the same basic idea of preconditions and effects, but apply it to agent skills and combine it with multiparty communication protocols. The resulting checker asks whether the declared goal can be achieved given both the available capabilities and the communication protocol.

Recent work on agent verification addresses related problems but has different goals. *Provable coordination via message sequence charts* [20] uses formal communication protocols for LLM agents and focuses on coordination correctness rather than goal achievability. Its runtime-planning extension also studies LLM-generated coordination protocols. *VeriGuard* [21] separates offline policy verification from online monitoring and uses the result as a safety mechanism. *Agent behavioural contracts* [22] provide probabilistic, runtime-enforced compliance and composition conditions for multi-agent systems. In contrast, our checker makes a static achievability decision and guarantees that an *impossible* verdict is correct for the formal pack.

*Skill-bundle scanners and verified-skill catalogs* address the same deployment object: portable agent skills. NVIDIA’s SkillSpector [23] and Verified Agent Skills pipeline [24] provide checks for skill bundles before deployment, including code, metadata, dependencies, and other security-related properties. These checks are complementary to our approach. They can inspect the original SKILL.md and its supporting files, while our checker operates on the formal pack produced from the skill. Their purpose is to identify security and consistency problems; ours is to determine whether the declared goal can be achieved with the available capabilities and protocol.

To our knowledge, the work above does not combine capability-based goal reachability with multiparty protocol checking for agent skills, nor provide the same refutation-sound guarantee for an *impossible* verdict.

## 7 Limitations and Future Work

- **Dependence on the declared pack.** The guarantees apply to the declared  $\Gamma$  and  $S$ . If the LLM incorrectly describes a capability or omits an important detail from the skill, the checker can only reason about the resulting pack. A practical system should therefore also inspect SKILL.md, scripts, dependencies, manifests, and permission metadata before running the achievability check.



245 • **Adding participants during execution.** The current procedure is decidable when the set of  
 246 participants is fixed. When new participants can be added during execution, the procedure  
 247 may return UNKNOWN rather than make an unsupported decision (Proposition 2).

248 • **Abstraction.** A coarse  $\alpha$  allows more behaviors to be treated as equivalent. This preserves  
 249 the refutation guarantee of Theorem 1, but may make the checker more likely to return  
 250 *achievable*.

251 • **Formalization gaps.** The Coq development covers the generic refutation-sound abstraction  
 252 theorem and concrete examples, head-move action safety, and subject reduction/session  
 253 fidelity with full bystander interleaving for the single-label communication model, all axiom-  
 254 free. The executable checker is schema-gated and tested against the formal specification, but  
 255 its exact symbolic state is not yet fully mechanized. The Coq model also does not yet com-  
 256 bine observable goal labels, participant matching, multi-branch choice, and world-changing  
 257 interleaving in one development. Future work includes completing this mechanization, con-  
 258 necting interleaved session runs to the reachability search, proving equivalence between the  
 259 declarative judgment and the projection-based adapter, and proving that abstract witnesses  
 260 correspond to concrete sessions. The 21-pack evaluation is also only a proof of concept.

261 **Broader impact.** Pre-execution checking can reduce wasted computation (Appendix E) and prevent  
 262 structurally impossible plans from reaching execution. However, an incorrect pack can create false  
 263 confidence, for example by omitting a tool effect or weakening the user’s actual goal. We therefore  
 264 report the formal pack, the result of the check, and any remaining conditions that still require attention.  
 265 Human review and runtime checks are still needed for consequential use. This work does not evaluate  
 266 fairness, privacy, or deployment safety.

## 267 8 Conclusion

268 Agent skills provide instructions, workflows, and tools that guide autonomous agents toward particular  
 269 outcomes. Whether a skill is written by a human expert or generated with the help of an LLM, a basic  
 270 question remains: **can the declared goal be achieved at all?** Our work answers this question before  
 271 execution, using a deterministic checker over a formal pack derived from the skill. The checker  
 272 combines capability-based reachability with a directly checked global protocol, and its *impossible*  
 273 verdict is sound for the formal pack it receives.

274 This guarantee does not remove the gap between the original skill and its formal representation. An  
 275 incorrect or incomplete pack can lead to an incorrect *achievable* result, which is why the result must  
 276 be understood relative to the declared pack. Nevertheless, when the pack is correct, the check can  
 277 identify structural failures before the agent spends time and tokens executing the skill. This provides  
 278 a simple and inspectable way to check whether an agent skill is achievable before execution.

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## A Syntax

We use disjoint sets of predicates  $p \in \text{Pred}$  (boolean), numeric variables  $x \in \text{Var}$ , roles  $p, q, r \in \mathcal{R}$ , capability names  $a \in \text{Cap}$ , and message labels  $\ell \in \text{Lab}$ . Following the modern presentation of multiparty sessions [1, 2] we define the model from the bottom up: state and capabilities (Sections A.1–A.2), then *processes* (Section A.3), and finally the global *type* they are checked against directly (Section A.4).

In each grammar, “ $::=$ ” lists the possible forms of an expression. In each inference rule below, the statement below the line holds when all premises above the line hold. Rule names identify the corresponding checker obligation; they are not additional assumptions.

### A.1 State logic

$$\begin{aligned} e &::= x \mid n \mid e + e \mid e - e \mid c \cdot e & (c, n \in \mathbb{Z}) \\ \varphi &::= p \mid \top \mid \perp \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid e \bowtie e & \bowtie \in \{<, \leq, =, \geq, >, \neq\} \end{aligned}$$

The state logic  $\mathcal{L}$  is quantifier-free linear integer arithmetic [19] with uninterpreted boolean predicates — a decidable fragment for which entailment  $\models$  and satisfiability are discharged by a satisfiability-modulo-theories (SMT) solver.



## 339 A.2 Worlds and capabilities

340 A world  $W = \langle B, N \rangle$  assigns truth values to predicates  $B : \text{Pred} \rightarrow \mathbb{B}$  and numeric variables  
341  $N : \text{Var} \rightarrow \mathbb{Z}$ . A *capability context*  $\Gamma$  maps each available tool  $a$  to a guarded effect:

$$\begin{aligned} \Gamma(a) &= \langle pre_a, eff_a \rangle, & pre_a &\in \mathcal{L}, \\ eff_a &= \langle Add_a \subseteq \text{Pred}, Del_a \subseteq \text{Pred}, Asg_a : \text{Var} \rightarrow e, Nd_a : \text{Var} \rightarrow \mathcal{L} \rangle \end{aligned}$$

342  $a \notin \text{dom}(\Gamma) \iff$  the agent does *not* possess tool  $a$  (no hallucinated tools).

343 *Add/Del* are the STRIPS add- and delete-lists; *Asg* gives deterministic numeric updates  $x := e$ ; *Nd*  
344 gives *nondeterministic* updates  $x := *$  constrained by a formula over the new value (used to model  
345 post-conditions such as “books a fare under 500”).

## 346 A.3 Processes and multiparty sessions

347 Skills *execute* as processes; a multi-agent run is a session of named processes. We use the synchronous  
348 multiparty session calculus in its modern coinductive **presentation [1, 2]: processes coinductively**  
349 **defined are possibly infinite regular trees (finitely many distinct subtrees)**, and labels are goal-relevant  
350 tokens,  $\ell = \alpha(m)$  for concrete messages  $m$  — the tolerance dial  $\alpha$  is a parameter of the label  
351 alphabet.

$$P ::=^{coind} \mathbf{0} \mid p!\{\ell_i.P_i\}_{i \in I} \mid p?\{\ell_i.P_i\}_{i \in I} \mid a.P \quad \mathbb{M} = p_1[P_1] \parallel \cdots \parallel p_n[P_n]$$

352 with  $I$  finite and non-empty, labels  $\ell_i$  pairwise distinct, and **role** names pairwise distinct. The output  
353  $p!\{\ell_i.P_i\}$  *internally* chooses a label and sends it to  $p$ ; the input  $p?\{\ell_i.P_i\}$  offers an *external* choice;  
354  $a.P$  invokes capability  $a$  — the only construct that changes the world. A skill  $S_p$  is a process: the  
355 behaviour declared for role  $p$ .

## 356 A.4 Global types

357 The single communication construct of a global type is a *directed* labelled choice — selection by  $p$   
358 and branching at  $q$  in one construct. (A partnerless “ $p$  selects” admits no coherent view for the roles  
359 that did not choose; directed choice is the standard MPST construct [13, 9].)

$$G ::=^{coind} p \rightarrow q : \{\ell_i.G_i\}_{i \in I} \mid a@p.G \mid \check{\varphi}.G \mid \text{End}$$

360 subject to  $p \neq q$  and the same regularity and label conditions as processes. This is the *only* type  
361 in the discipline: Section C.2 types a whole session directly against  $G$  — there is no local-type  
362 grammar, no projection function, and no separate subtyping relation. Goal markers  $\check{\varphi}$  live *only*  
363 in  $G$ : achievability is decided on the global type (Rule ACH in C.3), and markers are consumed at  
364 typing time by T-GOAL without ever burdening a process. Only  $a@p$  changes the world; messages  
365 carry coordination information only.

## 366 B Operational Semantics

### 367 B.1 World transitions

A capability fires when its guard holds, producing a successor world. Because *Nd* is nondeterministic,  
 $\langle W \rangle a \langle W' \rangle$  may relate one world to many. The relation is implicitly indexed by  $\Gamma$ : the notation  
 $pre_a, eff_a$  is defined only when  $\Gamma(a) = \langle pre_a, eff_a \rangle$ , so every firing presupposes  $a \in \text{dom}(\Gamma)$ .

$$\frac{\text{WORLD-ACT} \quad W \models pre_a \quad W' \in \text{apply}(eff_a, W)}{\langle W \rangle a \langle W' \rangle}$$

368  $\text{apply}(eff_a, \langle B, N \rangle) = \langle B', N' \rangle, \quad B' = (B \cup Add_a) \setminus Del_a$

$$N'(x) = \begin{cases} \llbracket Asg_a(x) \rrbracket_N & \text{if } x \in \text{dom}(Asg_a) \\ v \text{ with } \langle B, N[x \mapsto v] \rangle \models Nd_a(x) & \text{if } x \in \text{dom}(Nd_a) \\ N(x) & \text{otherwise (frame)} \end{cases}$$

## 370 B.2 A labelled semantics for sessions and global types

371 Only capabilities touch the world, so sessions reduce as configurations  $(\mathbb{M}; W)$ ; communication is  
 372 synchronous [10]. Each transition carries a label  $\Lambda$  recording its *participants*, so that independent  
 373 moves by disjoint roles can be identified. Two auxiliary maps drive the labelled system: the  
 374 *participants*  $\text{prt}(\cdot)$  of a global type, a label, or a session, and the *capabilities*  $\text{cap}(\cdot)$  (the set of moves  
 375 a global type can perform). On global types,

$$\begin{aligned} \text{prt}(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}) &= \{p, q\} \cup \bigcup_{i \in I} \text{prt}(G_i) & \text{prt}(\check{\varphi}. G) &= \text{prt}(G) \\ \text{prt}(a @ p. G) &= \{p\} \cup \text{prt}(G) & \text{prt}(\text{End}) &= \emptyset, \end{aligned}$$

376 on labels  $\text{prt}(p \ell q) = \{p, q\}$ ,  $\text{prt}(a @ p) = \{p\}$ ,  $\text{prt}(\check{\varphi}) = \emptyset$ , and on sessions  $\text{prt}(\mathbb{M}) = \{p \mid \mathbb{M} \equiv$   
 377  $p[P] \ \& \ P \neq \mathbf{0}\}$  (the roles with a residual behaviour). The capabilities of a global type are

$$\begin{aligned} \text{cap}(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}) &= \{p \ell_i q \mid i \in I\} \cup \bigcup_{i \in I} \text{cap}(G_i) \\ \text{cap}(a @ p. G) &= \{a @ p\} \cup \text{cap}(G) & \text{cap}(\check{\varphi}. G) &= \{\check{\varphi}\} \cup \text{cap}(G) & \text{cap}(\text{End}) &= \emptyset. \end{aligned}$$

378 The capability names used by a protocol are

$$\text{caps}(G) = \{a \mid \exists p. a @ p \in \text{cap}(G)\}.$$

The LTS of sessions is defined by

S-COMM

$$\frac{k \in I \subseteq J}{(p[q!\{\ell_i. P_i\}_{i \in I}] \parallel q[p?\{\ell_j. Q_j\}_{j \in J}] \parallel \mathbb{M}; W) \xrightarrow{p \ell_k q} (p[P_k] \parallel q[Q_k] \parallel \mathbb{M}; W)}$$

S-ACT

$$\frac{\langle W \rangle a \langle W' \rangle}{(p[a.P] \parallel \mathbb{M}; W) \xrightarrow{a @ p} (p[P] \parallel \mathbb{M}; W')}$$

S-GOAL

$$\frac{W \models \varphi}{(\mathbb{M}; W) \xrightarrow{\check{\varphi}} (\mathbb{M}; W)}$$

379 where  $\parallel$  is commutative, associative and  $p[\mathbf{0}] \parallel \mathbb{M} \equiv \mathbb{M}$ . A non-terminated configuration with  
 380 no successor is *stuck*; the deadlocking planner/worker handoff of Section 1 is the stuck two-role  
 381 configuration in which both processes begin with an input. A goal marker is *not* carried by any  
 382 process; instead S-GOAL lets a configuration *observe* that its world already entails one of its declared  
 383 goals  $\varphi$ , emitting the label  $\check{\varphi}$  without altering  $\mathbb{M}$  or  $W$ . (Each session carries a set of goal formulas,  
 384 and  $\varphi$  ranges over that set — not over every formula the world happens to satisfy — so that each  
 385 observation has a matching marker in the protocol.) This observable goal step is matched at the  
 386 type level by G-GOAL-E below, so that  $\check{\varphi}$  is a genuine label  $\Lambda$  of both transition systems and the  
 387 operational correspondence of Section D.3 ranges over it uniformly.

The checker relates a session to its protocol through a *labelled* transition system on global types,  
 $(G, W) \rightsquigarrow_{\Lambda} (G', W')$ , built from *head* rules that consume the leading construct and *interleaving*  
 rules that let a move by participants *disjoint* from the head commute inward:

G-COMM-E

$$\frac{k \in I}{(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}, W) \rightsquigarrow_{p \ell_k q} (G_k, W)}$$

G-ACT-E

$$\frac{\langle W \rangle a \langle W' \rangle}{(a @ p. G, W) \rightsquigarrow_{a @ p} (G, W')}$$

G-GOAL-E

$$\frac{W \models \varphi}{(\check{\varphi}. G, W) \rightsquigarrow_{\check{\varphi}} (G, W)}$$

G-COMM-I

$$\frac{(G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W') \quad \Lambda \in \text{cap}(G_i) \ \forall i \in I \quad \text{prt}(\Lambda) \cap \{p, q\} = \emptyset}{(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}, W) \rightsquigarrow_{\Lambda} (p \rightarrow q : \{\ell_i. G'_i\}_{i \in I}, W')}$$

G-ACT-I

$$\frac{(G, W) \rightsquigarrow_{\Lambda} (G', W') \quad \Lambda \in \text{cap}(G) \quad \text{prt}(\Lambda) \cap \{p\} = \emptyset}{(a @ p. G, W) \rightsquigarrow_{\Lambda} (a @ p. G', W')}$$

G-GOAL-I

$$\frac{(G, W) \rightsquigarrow_{\Lambda} (G', W') \quad \Lambda \in \text{cap}(G)}{(\check{\varphi}. G, W) \rightsquigarrow_{\Lambda} (\check{\varphi}. G', W')}$$

388 G-GOAL-E discharges a goal marker when  $W \models \varphi$ , emitting the observable label  $\checkmark \varphi$  that matches  
 389 S-GOAL. In the head-only reduction used for achievability, an unsatisfied marker is a checkpoint and  
 390 blocks the path. In the full labelled correspondence, G-GOAL-I lets a continuation step commute  
 391 inward under a still-pending marker. When that step changes the world, the correspondence theorem  
 392 uses the goal-stability hypothesis stated in Section D.3; removing that hypothesis is future work. The  
 393 extraction (head) rules G-COMM-E/G-ACT-E/G-GOAL-E give the  $\Lambda$ -erased reduction  $(G, W) \rightsquigarrow$   
 394  $(G', W')$  that the achievability judgment below uses; the interleaving rules matter only for the  
 395 operational correspondence of Section D.3, and their  $\Lambda \in \text{cap}(G)$  premise confines a commuting  
 396 move to a capability the residual protocol actually offers. G-COMM-E is legitimate because choice  
 397 is directed — the branch is a communication both endpoints observe, and Section C.2 checks that  
 398 every third party can follow. A well-formed pack uses its declared goal  $\varphi_{\text{goal}}$  at each goal marker. A  
 399 configuration is *goal-satisfying* when control is at such a marker or at the terminus and the world  
 400 entails that declared goal:

$$401 \quad \begin{aligned} & \text{goal}_{\varphi_{\text{goal}}}(\checkmark \varphi_{\text{goal}}.G, W) \iff W \models \varphi_{\text{goal}}, \quad \text{goal}_{\varphi_{\text{goal}}}(\text{End}, W) \iff W \models \varphi_{\text{goal}}, \\ & \Gamma; G \models \Diamond_{\text{init}} \varphi_{\text{goal}} \iff \exists W_0 \models \text{init}. \exists (G', W'). (G, W_0) \rightsquigarrow^* (G', W') \wedge \text{goal}_{\varphi_{\text{goal}}}(G', W'). \end{aligned}$$

402 The diamond is *existential*: a single witnessing run suffices — the formal source of detour-tolerance.

### 403 B.3 Realizability and subtyping, without projection

404 The classical route to ruling out a deadlocking handoff is *projection*: compute each role’s local  
 405 type  $G|r$  by structural recursion on  $G$ , taking the *merge* of a choice’s branches at every role that  
 406 neither sends nor receives, and declaring  $G$  *realizable* exactly when this partial function is defined  
 407 everywhere — undefinedness of the merge is the static signature of a role forced to behave differently  
 408 in branches it cannot distinguish. Session subtyping  $\leq$  is then a second, separate coinductive relation  
 409 on local types, letting a role’s actual behaviour offer more external choices and fewer internal ones  
 410 than its projected contract demands [14, 10, 11].

411 Both devices exist to answer questions about the *network as a whole* — “can every role tell the  
 412 branches apart it needs to?”, “may this implementation stand in for that contract?” — while type-  
 413 checking only one role’s syntax at a time. Section C.2 answers the same two questions with a single  
 414 judgment, *typing the whole configuration directly against  $G$* : a choice node’s rule quantifies over  
 415 every branch the protocol declares and requires the *same* residual configuration to check against each  
 416 one. This is the conservative plain-merge condition, obtained without ever computing  $\sqcap$  or a local  
 417 type. The receiver-side subtyping direction (a receiver may accept a superset of the protocol’s labels)  
 418 is folded into the same rule as a label-set inclusion  $I \subseteq J$ . Tolerance therefore still has the same  
 419 three independent knobs from Section 2 — the existential  $\Diamond$  (Section B.2), subtyping (now inside  
 420 T-COMM, Section C.2), and the granularity of  $\alpha$  — “flexible, tolerant of small details” is still the  
 421 profile  $\langle \text{coarse } \alpha, \Diamond, \text{linear } \mathcal{L} \rangle$ ; a later safety layer flips the same machinery to the universal  $\Box$  over  
 422 permission predicates.

## 423 C The Achievability Type System

424 Achievability now factors into three premises, each separately decidable, combined by the top rule  
 425 ACH: no hallucinated capability, direct conformance of the declared skills to  $G$ , and reachability of  
 426 the goal.

### 427 C.1 Capability and goal typing

$$\begin{array}{c} \text{T-EFF} \\ \frac{\Gamma(a) = \langle \text{pre}, \text{eff} \rangle \quad \varphi \models \text{pre}}{\Gamma \vdash \{\varphi\} a \{\text{sp}(\varphi, \text{eff})\}} \end{array} \quad \begin{array}{c} \text{T-MISS} \\ \frac{a \notin \text{dom}(\Gamma)}{\Gamma \vdash a \triangleright \mathbf{X}(\text{MISSING\_CAPABILITY})} \end{array}$$

428  $\text{sp}(\varphi, \text{eff})$  is the strongest postcondition of the STRIPS effect. T-MISS fires on hallucinated planning:  
 429 the plan names a tool the context does not grant, and achievability is refuted without any search.

### 430 C.2 Direct conformance typing

Skills are typed *directly* against the global protocol and the world, by the coinductive judgment  $G \vdash$   
 $(\mathbb{M}; W)$  — read “ $G$  types session  $\mathbb{M}$  starting from world  $W$ .” There is no local type, no projection,

and no separate subtyping relation: the receiver-side subtyping direction and the unobserved-choice check are structural side conditions of one rule.

$$\begin{array}{c}
\text{T-END} \\
\hline
\text{End} \vdash (p[0]; W)
\end{array}
\quad
\begin{array}{c}
\text{T-ACT} \\
\hline
\frac{G \vdash (p[P] \parallel \mathbb{M}; W') \quad \langle W \rangle a \langle W' \rangle \quad \text{prt}(G) \cup \{p\} = \text{prt}(\mathbb{M}) \cup \{p\}}{a@p. G \vdash (p[a.P] \parallel \mathbb{M}; W)}
\end{array}$$

$$\begin{array}{c}
\text{T-GOAL} \\
\hline
\frac{G \vdash (\mathbb{M}; W) \quad W \models \varphi \quad \text{prt}(G) = \text{prt}(\mathbb{M})}{\checkmark \varphi. G \vdash (\mathbb{M}; W)}
\end{array}$$

$$\begin{array}{c}
\text{T-COMM} \\
\hline
\frac{G_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}; W) \quad \forall i \in I \subseteq J \quad \text{prt}(p \rightarrow q : \{\ell_i.G_i\}_{i \in I}) = \text{prt}(\mathbb{M}) \cup \{p, q\}}{p \rightarrow q : \{\ell_i.G_i\}_{i \in I} \vdash (p[q!\{\ell_i.P_i\}_{i \in I}] \parallel q[p?\{\ell_j.Q_j\}_{j \in J}] \parallel \mathbb{M}; W)}
\end{array}$$

431 T-ACT pairs type and world in lockstep with WORLD-ACT: the unconsumed type  $a@p.G$  sits with  
 432 the *pre*-effect world  $W$  and the residual  $G$  with the *post*-effect world  $W'$  — the same convention  
 433 as G-ACT-E (Section B.2). In T-COMM the sender  $p$  offers exactly the protocol's branch set  $I$ ,  
 434 while the receiver  $q$  may accept *more* ( $I \subseteq J$ ) — the one subtyping direction the rule keeps (safe  
 435 interface slack at the receiver). Requiring *every* protocol branch  $\ell_i$  to be checked against the *same*  
 436 bystanders  $\mathbb{M}$  is the plain-merge form of the unobserved-choice condition, so a role forced to behave  
 437 differently across branches it cannot distinguish makes no derivation possible — the rule fails closed,  
 438 and deadlock-freedom falls out of T-COMM itself with no merge to compute (Theorem 4). Each  
 439 rule additionally carries a *participant-matching* side condition tying the protocol's participants to  
 440 the session's:  $\text{prt}(p \rightarrow q : \{\ell_i.G_i\}) = \text{prt}(\mathbb{M}) \cup \{p, q\}$  in T-COMM,  $\text{prt}(G) \cup \{p\} = \text{prt}(\mathbb{M}) \cup \{p\}$   
 441 in T-ACT, and  $\text{prt}(G) = \text{prt}(\mathbb{M})$  in T-GOAL. These make the invariant “the roles the protocol still  
 442 mentions are exactly the roles still running” hold at every node, ruling out a session that carries a stray  
 443 participant the protocol has forgotten (or vice versa) — the well-formedness that lets the interleaving  
 444 cases of Section D.3 match a bystander move on the session to a  $\text{cap}(\cdot)$ -labelled move on the type.  
 445 T-END closes the discipline at the terminus: the terminated protocol  $\text{End}$  types a single idle role  $p[0]$   
 446 — and, through the congruence  $p[0] \parallel \mathbb{M} \equiv \mathbb{M}$ , any session all of whose roles have terminated —  
 447 the base case of the participant invariant, since  $\text{prt}(\text{End}) = \emptyset$  and an idle role contributes nothing to  
 448  $\text{prt}(\mathbb{M})$ . T-ACT is derivable only when  $\langle W \rangle a \langle W' \rangle$  holds, which itself presupposes  $a \in \text{dom}(\Gamma)$ :  
 449 this is where hallucinated capabilities die, with T-MISS as the diagnostic.

### 450 C.3 The achievability judgment

$$\begin{array}{c}
\text{ACH} \\
\hline
\frac{\text{caps}(G) \subseteq \text{dom}(\Gamma) \quad \exists W_0 \models \text{init}. G \vdash (p_1[S_{p_1}] \parallel \dots \parallel p_n[S_{p_n}]; W_0) \wedge \Gamma; (G, W_0) \models \Diamond \varphi_{\text{goal}}}{\Gamma \vdash \{S_p\}_p : G \triangleright \Diamond \varphi_{\text{goal}}}
\end{array}$$

451 Read top-to-bottom: no hallucinated tools; some plausible initial world exists from which the declared  
 452 skills directly conform to the protocol (deadlock-free, every capability call guarded) *and* that same  
 453 world reaches the goal. The reachability half is unchanged from Section B.2 and decided on  $\Gamma, G$   
 454 alone, before any  $S_p$  is known — exactly what lets the checker run on the LLM-compacted pack  
 455 the moment it exists; conformance is the additional, separately decidable check once the skills are  
 456 declared, and needs no transport lemma: the receiver-side subtyping slack is already inside T-COMM.

### 457 C.4 Reachability rules

The diamond is derived by the following inductive system, realized by the checker as a finite forward search over  $\rightsquigarrow$ ; unlike T-COMM/T-ACT/ T-GOAL, it never mentions a session  $\mathbb{M}$ , which is what makes it decidable on the pack alone.

$$\begin{array}{c}
\text{REACH-GOAL} \\
\hline
\frac{W \models \varphi_{\text{goal}} \quad G = \text{End} \vee \exists G'. G = \checkmark \varphi_{\text{goal}}. G'}{\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}}
\end{array}$$

$$\begin{array}{c}
\text{REACH-STEP} \\
\hline
\frac{(G, W) \rightsquigarrow (G', W') \quad \Gamma; (G', W') \models \Diamond \varphi_{\text{goal}}}{\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}}
\end{array}$$

$$\begin{array}{c}
\text{REACH-OR} \\
\hline
\frac{\exists k \in I. \Gamma; (G_k, W) \models \Diamond \varphi_{\text{goal}}}{\Gamma; (p \rightarrow q : \{\ell_i.G_i\}_{i \in I}, W) \models \Diamond \varphi_{\text{goal}}}
\end{array}$$

## 458 D Metatheory

459 We separate what is *mechanized in Coq* (Theorems 1–3; Theorems 4–5 for the single-label com-  
 460 munication fragment with full interleaving; the head-move action case; and the concrete instances,  
 461 all axiom-free) from what is *proved on paper* (the action-interleaving cases, decidability, and the  
 462 undecidability boundary).

### 463 D.1 Soundness of the abstraction

464 Let  $\mathcal{C}$  be the concrete configuration space of protocol/world pairs,  $\mathcal{A}$  the abstract configuration space  
 465 the checker explores, and  $abs : \mathcal{C} \rightarrow \mathcal{A}$  the abstraction. We require:

$$(step\text{-}sim) \ C \rightsquigarrow C' \Rightarrow abs(C) \hat{\rightsquigarrow} abs(C'), \quad (goal\text{-}sim) \ goal_{\varphi_{goal}}(C) \Rightarrow \hat{goal}(abs(C)).$$

466 **Lemma 1** (Transport). *If  $C_0 \rightsquigarrow^* C$  then  $abs(C_0) \hat{\rightsquigarrow}^* abs(C)$ .*

467 *Proof.* By induction on the reflexive-transitive closure. The empty run is reflexivity. For  $C_0 \rightsquigarrow^*$   
 468  $C' \rightsquigarrow C$ , the induction hypothesis gives  $abs(C_0) \hat{\rightsquigarrow}^* abs(C')$  and STEP-SIM gives  $abs(C') \hat{\rightsquigarrow} abs(C)$ ;  
 469 compose them. Mechanized as `reach_abs`, parameterized by STEP-SIM.  $\square$

470 **Lemma 2** (Reachability monotonicity). *If every step of  $\rightsquigarrow_1$  is a step of  $\rightsquigarrow_2$  (that is,  $x \rightsquigarrow_1 y$  implies  
 471  $x \rightsquigarrow_2 y$ ), then  $s \rightsquigarrow_1^* w$  implies  $s \rightsquigarrow_2^* w$ .*

472 *Proof.* By induction on the closure  $s \rightsquigarrow_1^* w$ . The empty run transports by reflexivity. For  $s \rightsquigarrow_1^* u \rightsquigarrow_1$   
 473  $w$ , the induction hypothesis gives  $s \rightsquigarrow_2^* u$ ; the hypothesis turns the last step  $u \rightsquigarrow_1 w$  into  $u \rightsquigarrow_2 w$ ;  
 474 appending it gives  $s \rightsquigarrow_2^* w$ . Mechanized as `reach_mono`.  $\square$

### 475 D.2 Refutation soundness, tolerance, monotonicity

476 **Theorem 1** (Refutation soundness). *If the abstract system has no reachable goal state from  
 477  $abs(G, W_0)$ , then no declared run of that pack configuration reaches  $\varphi_{goal}$ . Hence an impossi-  
 478 ble verdict is never wrong relative to the pack, provided STEP-SIM and GOAL-SIM hold.*

479 *Proof.* We prove the contrapositive: if *some* declared run reaches the goal, then the abstract  
 480 system reaches an abstract goal state. Suppose  $C_0 \rightsquigarrow^* C$  with  $goal_{\varphi_{goal}}(C)$ . By Lemma 1,  
 481  $abs(C_0) \hat{\rightsquigarrow}^* abs(C)$ , and by GOAL-SIM,  $\hat{goal}(abs(C))$ . Thus  $abs(C)$  is an abstract goal state reach-  
 482 able from  $abs(C_0)$ , contradicting the hypothesis. Hence no declared run reaches  $\varphi_{goal}$ : a verdict  
 483 of *impossible* — read off exactly the premise, that the abstract search exhausts without hitting an  
 484 abstract goal — is never wrong. Mechanized, axiom-free as a theorem schema parameterized by the  
 485 two simulation hypotheses (the existential witness is  $abs(C)$ ):  $\square$

```
486 Theorem refutation_sound : forall s0,
487   (~ exists a, reach_astype (abs s0) a /\ agoal a) ->
488   (~ exists w, reach_cstep s0 w /\ cgoal w).
489   (* Print Assumptions refutation_sound. => Closed under the global
490   context *)
```

491 **Theorem 2** (Tolerance soundness). *If a coarser over-approximation (more edges) cannot reach the  
 492 goal, neither can any finer system. Hence coarsening  $\alpha$  to tolerate more detail never produces a false  
 493 refutation.*

494 *Proof.* Let  $\rightsquigarrow_{fine}$  and  $\rightsquigarrow_{coarse}$  be the two abstract step relations, with the coarsening an *over-*  
 495 approximation: every fine step is also a coarse step,  $x \rightsquigarrow_{fine} y \Rightarrow x \rightsquigarrow_{coarse} y$ . Suppose, for  
 496 contradiction, that the finer system reaches the goal:  $s_0 \rightsquigarrow_{fine}^* a$  with  $\hat{goal}(a)$ . By Lemma 2 (with  
 497  $\rightsquigarrow_1 = \rightsquigarrow_{fine}$ ,  $\rightsquigarrow_2 = \rightsquigarrow_{coarse}$ ),  $s_0 \rightsquigarrow_{coarse}^* a$ ; and  $\hat{goal}(a)$  still holds. So the coarser system reaches  
 498 a goal state, contradicting the hypothesis. Hence no coarser-refuted goal is reachable in any finer  
 499 system: coarsening  $\alpha$  only *adds* abstract edges, so it can never turn an unreachable goal into a  
 500 reachable one — refutation is preserved. Mechanized as `tolerance_sound`, axiom-free.  $\square$

**Theorem 3** (Capability monotonicity). *If  $\Gamma \subseteq \Gamma'$  and  $\Gamma$  reaches  $\varphi_{goal}$ , then  $\Gamma'$  reaches  $\varphi_{goal}$ . Granting tools never makes an achievable goal impossible.*

*Proof.* Enlarging the capability context only *adds* guarded effects: since  $\Gamma \subseteq \Gamma'$ , every capability firing available under  $\Gamma$  is available under  $\Gamma'$  with the same guard and effect (WORLD-ACT in B.1 quantifies over  $a \in \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$ ), so every step of the  $\Gamma$ -transition relation is a step of the  $\Gamma'$ -relation. Suppose  $\Gamma$  reaches the goal configuration  $C$  from  $C_0$ . By Lemma 2, the same configuration is reachable under  $\Gamma'$ , and  $\text{goal}_{\varphi_{goal}}(C)$  is unchanged because the goal predicate does not depend on  $\Gamma$ . Hence  $\Gamma'$  reaches  $\varphi_{goal}$  via the same witnessing run. The other premises of ACH are monotone in the same direction ( $\text{caps}(G) \subseteq \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$  still holds, and realizability and conformance do not mention  $\Gamma$  beyond  $a \in \text{dom}(\Gamma)$ ), so the whole judgment is preserved. The transport half is mechanized as `cap_monotone`, axiom-free, stated over an inclusion of step relations; the reduction of  $\Gamma \subseteq \Gamma'$  to that inclusion is the WORLD-ACT argument above, on paper.  $\square$

**Concrete instance.** The flight skill of Section 2, variant A, is mechanized as `FlightInstance`. We prove `flight_refuted` (no run reaches `booked`  $\wedge$  `confirmation_sent`, since no step touches `conf`) and `booking_reachable` (a booking is still reachable). Both are axiom-free, witnessing non-vacuity of Theorem 1.

### D.3 Operational correspondence: subject reduction and session fidelity

The direct judgment  $G \vdash (\mathbb{M}; W)$  is not merely preserved by reduction: the session and its protocol step *in lockstep* over the labelled transition systems of Section B.2. This is the standard soundness backbone of a session calculus [13, 11], obtained here *directly*, with no projection and no merge. For world-changing interleavings, the statements below assume two explicit frame conditions: effects of participant-disjoint actions commute, and an action moved under a pending marker preserves that marker's formula. Communication-only reductions satisfy both conditions vacuously.

**Lemma 3** (Residual capability). *If  $(G, W) \rightsquigarrow_{\Lambda} (G', W')$ , then  $\Lambda \in \text{cap}(G)$ .*

**Lemma 4** (Participant agreement). *If  $G \vdash (\mathbb{M}; W)$ , then  $\text{prt}(G) = \text{prt}(\mathbb{M})$ .*

Both follow by induction on the transition or typing derivation, respectively; they are the capability-membership and participant invariants used in the interleaving cases below.

**Theorem 4** (Subject reduction). *If  $G \vdash (\mathbb{M}; W)$  and  $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$ , then  $(G, W) \rightsquigarrow_{\Lambda} (G', W')$  for some  $G'$  with  $G' \vdash (\mathbb{M}'; W')$ .*

**Theorem 5** (Session fidelity). *If  $G \vdash (\mathbb{M}; W)$  and  $(G, W) \rightsquigarrow_{\Lambda} (G', W')$ , then  $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$  for some  $\mathbb{M}'$  and  $W'$  with  $G' \vdash (\mathbb{M}'; W')$ .*

We prove both by induction on the typing derivation  $G \vdash (\mathbb{M}; W)$ , with a case analysis on the transition; the one auxiliary fact used throughout is that typing respects pointwise-equal sessions (`ctypes_ext`), which lets the interleaving cases reassociate independent updates.

*Proof of Theorem 4 (subject reduction).* By induction on  $G \vdash (\mathbb{M}; W)$ .

**Case T-END** ( $G = \text{End}$ , every role idle). Then  $\mathbb{M}$  has no output or pending action, so the only available transition is the goal observation S-GOAL at  $\Lambda = \checkmark \varphi_{goal}$  when  $W \models \varphi_{goal}$ , which leaves  $(\mathbb{M}, W)$  fixed; there is no residual type to preserve and the claim holds (otherwise no  $\Lambda$ -transition exists and it holds vacuously).

**Case T-GOAL** ( $G = \checkmark \varphi.G_0$ , with  $W \models \varphi$  and  $G_0 \vdash (\mathbb{M}; W)$ ,  $\text{prt}(G_0) = \text{prt}(\mathbb{M})$ ). The transition  $(\mathbb{M}, W) \rightarrow_{\Lambda} (\mathbb{M}', W')$  splits on its label.

- **Goal observation** ( $\Lambda = \checkmark \varphi$ , by S-GOAL, so  $\mathbb{M}' = \mathbb{M}$  and  $W' = W$ ). Since  $W \models \varphi$ , rule G-GOAL-E discharges the marker,  $(\checkmark \varphi.G_0, W) \rightsquigarrow_{\checkmark \varphi} (G_0, W)$ , and  $G_0 \vdash (\mathbb{M}; W)$  is the premise: take  $G' = G_0$ .
- **Interleaving** ( $\Lambda \neq \checkmark \varphi$ ). A goal marker is not a session construct, so this is already a transition of  $G_0$ 's session; the induction hypothesis gives  $(G_0, W) \rightsquigarrow_{\Lambda} (G', W')$  with



547  $\Lambda \in \text{cap}(G_0)$  and  $G' \vdash (\mathbb{M}'; W')$ . When the move is world-preserving (a communica-  
 548 tion,  $W' = W$ ), G-GOAL-I carries the still-pending marker along,  $(\check{\varphi}.G_0, W) \rightsquigarrow_{\Lambda}$   
 549  $(\check{\varphi}.G', W)$ , and  $\check{\varphi}.G' \vdash (\mathbb{M}'; W)$  by T-GOAL (the side condition  $\text{prt}(G') = \text{prt}(\mathbb{M}')$  is  
 550 preserved because  $\Lambda$  acts on the same participants on both sides). When the move is *world-*  
 551 *changing* (a capability firing,  $W' \neq W$ ), G-GOAL-I still commutes it under the marker,  
 552  $(\check{\varphi}.G_0, W) \rightsquigarrow_{\Lambda} (\check{\varphi}.G', W')$ ; re-typing the residual by T-GOAL then additionally asks  
 553  $W' \models \varphi$ . This follows from the goal-stability hypothesis on actions commuting under pend-  
 554 ing markers. The global type sequences the marker before  $G_0$ 's actions whereas the session  
 555 may fire the action first; the interleaving rule lets the type catch up under that hypothesis.  
 556 Removing the hypothesis, rather than this conditional case, is the open correspondence  
 557 problem recorded in Section 7.

558 *Case T-COMM*, with head  $p \rightarrow q$  and branches  $G_i$ . Here  $\mathbb{M}$  is  $p[q! \dots] \parallel q[p? \dots] \parallel \mathbb{M}_0$ , and each  
 559 branch satisfies  $G_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0; W)$ . If the transition is a goal observation  $\Lambda = \check{\varphi}$   
 560 (S-GOAL, leaving  $(\mathbb{M}, W)$  fixed), it is matched by G-COMM-I:  $\text{prt}(\check{\varphi}) = \emptyset$  makes the disjointness  
 561 premise trivial, and each branch realises the same observation by the induction hypothesis, using  
 562  $\check{\varphi} \in \text{cap}(G_i)$  — the observation concerns a marker the protocol actually pends (see the note below).  
 563 Otherwise the transition is a communication  $\Lambda = r \ell t$ . Because  $p$  offers only outputs to  $q$  and  $q$  only  
 564 inputs from  $p$ , the sender/receiver shapes force a dichotomy.

- 565 • *Head move* ( $r = p$ ). Then  $t = q$  and  $\ell = \ell_k \in I$ , and  $\mathbb{M}' = p[P_k] \parallel q[Q_k] \parallel \mathbb{M}_0$ . Rule  
 566 G-COMM-E gives  $(G, W) \rightsquigarrow_{p\ell_k q} (G_k, W)$ , and  $G_k \vdash (\mathbb{M}'; W)$  is exactly the  $k$ -th premise.
- 567 • *Interleaving move* ( $r \neq p$ ). The shapes then force  $r, t \notin \{p, q\}$ :  $r$  is an output so  $r \neq q$ ,  
 568 and  $t$  is an input so  $t \neq p$ , while  $t = q$  would force  $r = p$ . Since  $r, t$  lie in the bystanders  
 569  $\mathbb{M}_0$ , the *same* communication is enabled in each branch's configuration  $p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0$   
 570 (the head updates do not touch  $r, t$ ). The induction hypothesis at each  $i$  yields  $G'_i$  with  
 571  $(G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W)$  and  $G'_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}'_0; W)$ , where  $\mathbb{M}'_0$  is  $\mathbb{M}_0$  after the  
 572 bystander step. As  $\text{prt}(\Lambda) = \{r, t\}$  is disjoint from  $\{p, q\}$ , rule G-COMM-I assembles  
 573  $(G, W) \rightsquigarrow_{\Lambda} (p \rightarrow q: \{\ell_i. G'_i\}, W)$ , and re-applying T-COMM (the sender/receiver at  $p, q$   
 574 are untouched, and each branch is retyped by the induction hypothesis, reassociating the  
 575 independent updates via `ctypes_ext`) gives the residual typing.

576 Communications leave the world fixed, so no independence hypothesis is needed. For a world-  
 577 changing action at the head (T-ACT), G-ACT-E matches and the residual  $G$  is typed at the *same*  
 578 post-effect world the rule reaches — this is exactly where the lockstep world pairing of the corrected  
 579 T-ACT is used; a goal observation at an action-typed configuration is matched by G-ACT-I exactly  
 580 as in the T-COMM case, since  $\text{prt}(\check{\varphi}) \cap \{p\} = \emptyset$  trivially. Interleaving *two* actions requires their  
 581 effects to commute ( $\text{eff}_a \circ \text{eff}_b = \text{eff}_b \circ \text{eff}_a$  for disjoint  $a, b$ ), a frame-independence side condition  
 582 that participant-disjointness alone does not supply; under it the T-ACT interleaving case goes through  
 583 identically.  $\square$

584 *Proof of Theorem 5 (session fidelity)*. By induction on  $G \vdash (\mathbb{M}; W)$ , inverting the global step  
 585  $(G, W) \rightsquigarrow_{\Lambda} (G', W')$ . T-END admits only the goal observation G-GOAL-E at  $\check{\varphi}_{\text{goal}}$ , matched  
 586 on the session by S-GOAL. For T-GOAL, the step is either G-GOAL-E — matched by S-GOAL  
 587 on the session, the marker discharged — or G-GOAL-I, whose premise  $(G_0, W) \rightsquigarrow_{\Lambda} (G'_0, W')$   
 588 the induction hypothesis realises as an underlying session step that carries the marker along. For  
 589 T-COMM, the global step is either G-COMM-E — and the prescribed head communication  $p \ell_k q$  is  
 590 enabled by S-COMM, landing in the  $k$ -th premise — or G-COMM-I with  $\text{prt}(\Lambda) \cap \{p, q\} = \emptyset$ ; then  
 591 the induction hypothesis on any branch produces a bystander communication, whose endpoints lie  
 592 outside  $\{p, q\}$ , so S-COMM fires it on  $\mathbb{M}$  and T-COMM retypes the result. The action cases mirror  
 593 those of Theorem 4.  $\square$

594 **Mechanization.** Both theorems are mechanized axiom-free for the single-label communication  
 595 fragment — the fragment in which bystander interleaving is unconditional — in `DirectTypingSR.v`,  
 596 over labelled session and global transition systems, with no projection, merge, or local type:

```
597 Theorem subject_reduction : forall W G s s' L,
598   ctypes W G s -> lstep L s s' ->
```

```

599     exists G', gstep W L G G' /\ ctypes W G' s'.
600     Theorem session_fidelity : forall W G G' s L,
601       ctypes W G s -> gstep W L G G' ->
602       exists s', lstep L s s' /\ ctypes W G' s'.
603     (* Print Assumptions => Closed under the global context (both). *)

```

604 The head-move case for world-changing *actions* is additionally mechanized in `DirectTyping.v`  
605 (type\_directed\_safety, also progress). Observable goal labels appear in neither Coq develop-  
606 ment: the mechanized label alphabet is communication-only, and its goal rule fuses marker discharge  
607 with a continuation step. Participant-matching side conditions, labelled action interleaving, and  
608 multi-branch choice are likewise absent. Completing a faithful mechanization of the paper rules  
609 remains work in Section 7.

610 **Concrete instance.** `HandoffInstance` mechanizes the planner/worker handoff of Section 1 on both  
611 sides. The *good* pairing — planner sends, worker delivers, goal checked — is proved `ctypes`-typed  
612 (`good_typed`) and its run is proved to reach a world satisfying the goal (`good_runs_to_goal`). The  
613 *bad* pairing — both roles start with an input, exactly the deadlock of Section 1 — is proved `Stuck`  
614 (`bad_stuck`), and the contrapositive of progress then gives, for free, that *no non-trivial global type*  
615 *can ever type it* (`bad_untypable`): the rule rejects the classical deadlocking handoff without any  
616 projection ever being computed.

#### 617 D.4 Incompleteness, by design

618 **Proposition 1** (Incompleteness).  $\Gamma; G \models \Diamond_{init} \varphi_{goal}$  may hold while no concrete runtime execution  
619 reaches  $\varphi_{goal}$ : the witnessing abstract path can be spurious when  $\alpha$  collapses a distinction that  
620 mattered. The system is sound for  $\times$  and incomplete for  $\checkmark$ .

621 This is the price of decidable, tolerant over-approximation; tightening  $\alpha$  trades tolerance for precision.  
622 The residue is confined to two edges, motivating the layered architecture:

- 623 • **Payload faithfulness** (bottom edge): a tool’s declared post-condition (“filters to fares  
624 < 500”) may be false — a runtime obligation for a deterministic monitor (Layer C), not a  
625 static one.
- 626 • **Intent fidelity** (top edge): the formalized goal may not capture what the user meant —  
627 irreducibly semantic, surfaced at the single compaction checkpoint for human review.

#### 628 D.5 Decidability and the autonomy boundary

629 **Theorem 6** (Decidability). *In the fragment with finitely many roles (static topology), regular global*  
630 *types represented by finite tail-recursive control graphs, decidable state logic  $\mathcal{L}$ , and  $\mathcal{L}$ -definable*  
631 *effects,  $\Gamma; G \models \Diamond_{init} \varphi_{goal}$  is decidable.*

632 *Proof.* The pack-level achievability judgment  $\Gamma; G \models \Diamond_{init} \varphi_{goal}$  (Section B.2) is an existential  
633 reachability query over configurations  $(G', W)$ ; we exhibit a terminating decision procedure and  
634 argue its completeness.

635 *The search space is finite.* A configuration has two components. (i) *Control*, the residual global  
636 type  $G'$  reachable from  $G$  by  $\rightsquigarrow$ . Since  $G$  is a regular tree (tail-recursion: finitely many distinct  
637 subtrees, Section A.3) and each  $\rightsquigarrow$  step either descends into a subtree (G-COMM-E, G-ACT-E) or  
638 erases a marker (G-GOAL-E), the set  $\{G' : (G, \_) \rightsquigarrow^* (G', \_)\}$  is a subset of the finitely many  
639 subtrees of  $G$ , hence finite. (ii) *Symbolic world*. The boolean part is a valuation  $B : \text{Pred} \rightarrow \mathbb{B}$   
640 over the finitely many predicates named in the pack, so there are  $2^{|\text{Pred}|}$  boolean states. The numeric  
641 part is summarized by an accumulated constraint  $\psi \in \mathcal{L}$  (a conjunction of the guards taken and  
642 the  $Nd$ -postconditions assumed); on a control-graph back edge the reference procedure applies the  
643 widening  $\psi \mapsto \top$ , forgetting accumulated numeric constraints. This one-element numeric summary  
644 is an over-approximation and, together with the finite boolean valuations, leaves only finitely many  
645 symbolic worlds. The product with finite control is therefore finite.

646 *Each edge is computable.* Firing a capability requires checking  $W \models pre_a$  and forming  $sp$ ; testing  
647 goal-satisfaction requires  $W \models \varphi$ . Both are entailment/satisfiability queries in quantifier-free linear  
648 integer arithmetic with uninterpreted predicates — a decidable theory, discharged by an SMT solver.

Branch selection (REACH-OR) is a finite disjunction. Existence of an initial world is decided by the same SMT query over *init*; each satisfying symbolic initial valuation seeds the finite search.

*Termination and correctness.* Forward search (breadth-first over  $\rightsquigarrow$ ) maintains a visited set of (control, symbolic-world) pairs; because that set is finite it adds each pair at most once, so the search halts. It answers ACHIEVABLE iff it enumerates a goal-satisfying configuration, and IMPOSSIBLE otherwise. Soundness of the IMPOSSIBLE verdict is Theorem 1; and widening only *enlarges* the reachable set (it weakens constraints, adding edges), so by Theorem 2 it never introduces a false refutation. Hence the procedure is a decision procedure for  $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ .  $\square$

**Proposition 2** (Undecidability in a dynamic-topology extension). *Consider the extension  $G^+$  of the grammar with spawn  $r.G$ , which creates fresh participants at run time. With an unbounded role population and dynamic channels, achievability in  $G^+$  is undecidable.*

*Proof sketch.* By reduction from the control-state reachability problem for communicating finite-state machines (CFSMs), which is undecidable [15]. Fix a CFSM instance: finite-control machines  $M_1, M_2$  exchanging messages over unbounded FIFO channels, and a target control state  $q_f$  of  $M_1$ ; deciding whether a run reaches  $q_f$  is undecidable.

We compute, from this instance, a skill whose goal is achievable iff the CFSM reaches  $q_f$ . Each machine  $M_i$  becomes a role whose process mirrors its control graph: a transition that sends message  $m$  spawns, at run time, a fresh *courier* participant carrying  $m$  (this is exactly the dynamic-spawning capability the theorem hypothesizes), and a transition that receives  $m$  synchronizes with the oldest outstanding courier for that channel, so the unbounded family of couriers realizes the unbounded FIFO. The required order can be made explicit by assigning sequence numbers through the numeric world state and guarding receives on the next sequence number; the *Asg/Nd* machinery of Section A.2 supplies those updates. A boolean predicate  $at_{q_f}$  is established by the single capability guarding  $M_1$ ’s entry into  $q_f$ , and we set  $\varphi_{goal} = at_{q_f}$ . By construction a run of the skill reaches a state satisfying  $\varphi_{goal}$  iff the CFSM has a run reaching  $q_f$ : the courier discipline then puts the skill’s reachable configurations in correspondence with the CFSM’s channel contents and control states. The construction is a computable map from CFSM instances to packs, so a decision procedure for achievability would decide CFSM control-state reachability — impossible. Hence achievability is undecidable once participants may be spawned dynamically. (The finiteness argument of Theorem 6 breaks at exactly the point this reduction exploits: an unbounded number of couriers makes the set of reachable controls infinite.)  $\square$

The two results isolate one concrete boundary: static finite topology admits a total decision procedure, while unbounded dynamic spawning does not. This is a boundary on topology, not a claim that all forms of agent autonomy are undecidable. The implementation returns UNKNOWN when a pack violates the static-topology side condition of Theorem 6, unless an independent capability or establisher refutation has already proved it impossible.

## D.6 The partition of obligations

**Corollary 1.** *The architecture separates “will the goal be achieved” into four obligations on disjoint substrates:*

| Concern                           | Where                      | Mechanism                                       |
|-----------------------------------|----------------------------|-------------------------------------------------|
| goal achievability                | static, decidable          | Coq specification; deterministic search; no LLM |
| per-step safety / least privilege | Layer B (future)           | same engine, $\square$ + permission tiers       |
| payload faithfulness              | Layer C (future), run-time | deterministic monitors with blame               |
| intent fidelity                   | top edge, synthesis        | one human checkpoint                            |

## D.7 Verifiable feedback beyond a scalar reward

The verdict is a heterogeneous verification signal rather than a scalar score. An IMPOSSIBLE result identifies which premise failed and carries a blocking frontier; ACHIEVABLE carries a witness path and explicitly retains the two deferred obligations; UNKNOWN is an abstention outside the decidable fragment, not a refutation. These structured outcomes can serve as cheap negative labels

for automated skill or prompt search without an LLM judge, while the intent-fidelity checkpoint keeps the irreducible semantic decision with a human. We treat such search as an application of the verifier, not as a contribution evaluated here.

## E Token economics of pre-execution refutation

The practical case for deciding achievability *before* execution is an economic one, so we state it in tokens rather than adjectives. Two asymmetries carry it. First, no model sits in the trusted path: the schema gate, projection, conformance and Z3 reachability spend **zero tokens**, and the deterministic front-end spends zero as well. Only the optional LLM compaction spends tokens, and it is paid *once per skill version*. Second, an agent turn re-sends its whole context: writing  $S$  for the harness prompt,  $K$  for the loaded skill,  $g$  for the mean tokens appended per turn and  $o$  for output per turn, a run of  $T$  turns reads  $T(S+K) + gT(T-1)/2$  input and  $To$  output tokens. Compaction is linear in the skill and paid once; a doomed run is *quadratic* in its turns and recurs on every invocation.

Runtime waste is necessarily a model — it prices a run that, if the refutation is correct, never happens — so we publish the model rather than a single number. Each refutation reason carries a (low, typical, high) band for how long an agent flails before that failure stops it, and how many agents are billed. With conservative defaults ( $S=2500$ ,  $g=700$ ,  $o=250$ ) and a median real consumer skill ( $K \approx 1.7K$  tokens), compaction costs  $\sim 2.8K$  tokens against the following avoided waste per prevented invocation:

| Verdict reason     | turns (lo–typ–hi) | waste/run | leverage |
|--------------------|-------------------|-----------|----------|
| MISSING_CAPABILITY | 3–8–14, 1 agent   | 50 240    | 18×      |
| GOAL_UNSAT         | 8–18–30, 1 agent  | 176 040   | 63×      |
| NON_CONFORMANT     | 6–15–30, 2 agents | 261 900   | 94×      |
| NON_PROJECTABLE    | 6–15–30, 2 agents | 261 900   | 94×      |
| BLOCKED_GUARD      | 12–25–50, 1 agent | 305 750   | 110×     |

The ordering is the robust part, and it is the expected one. `MISSING_CAPABILITY` is cheapest at run time: the agent calls a tool that is not there and learns immediately, and most of the cost is the improvisation that follows. `BLOCKED_GUARD` is dearest, because a precondition no run can satisfy is precisely the case an agent cannot learn from — it retries to the harness turn cap. The two multi-party reasons bill two contexts. `GOAL_UNSAT` additionally carries a cost the token bill does not show: a run whose goal conjunct has no establisher can terminate *believing* it succeeded. `DYNAMIC_TOPOLOGY` claims no savings at all, because an abstention prevents nothing.

Over the 15-spec corpus the seven structural refutations cost 12.2K tokens of compaction and avoid  $\sim 1.07M$  tokens per round of invocations (band 0.28M–3.15M):  $\sim 87\times$  leverage, or nothing at all on the deterministic path. Break-even therefore falls *inside* the first prevented run in every mode — between 0.9% and 5.5% of it. The honest denominator for a healthy skill, where the check buys nothing and still costs something, is 2.9% of one successful run over the corpus and 4.0% for a median real skill; with the deterministic front-end, zero. Prompt caching changes the price of the wasted tokens, not their number. Halving every waste default leaves leverage at  $9\times$ – $55\times$  and the ordering unchanged. The model, its defaults and their justification are in the artifact (`skillc.tokens`, `skillc.cost`); the Coq development costs no tokens and is amortized over every skill ever checked.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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738 Question: Does the paper discuss the limitations of the work performed by the authors?

739 Answer: [Yes]

740 Justification: The Limitations and Future Work section discusses relative soundness, static

741 topology, abstraction coarseness, mechanization gaps, and the proof-of-concept corpus.

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743 Question: For each theoretical result, does the paper provide the full set of assumptions and

744 a complete (and correct) proof?

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746 Justification: Appendix D states the simulation, stability, commutativity, finiteness, and

747 topology assumptions with proofs or proof sketches; the artifact checks the explicitly

748 identified Coq fragments.

749 **4. Experimental result reproducibility**

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751 perimental results of the paper to the extent that it affects the main claims and/or conclusions

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813 synthetic corpora, proof scope, limitations, and reproduction workflow.

814 **14. Crowdsourcing and research with human subjects**

815 Question: For crowdsourcing experiments and research with human subjects, does the paper  
816 include the full text of instructions given to participants and screenshots, if applicable, as  
817 well as details about compensation (if any)?

818 Answer: [N/A]

819 Justification: The research involves neither crowdsourcing nor human-subject experiments.

820 **15. Institutional review board (IRB) approvals or equivalent for research with human  
821 subjects**

822 Question: Does the paper describe potential risks incurred by study participants, whether  
823 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)  
824 approvals (or an equivalent approval/review based on the requirements of your country or  
825 institution) were obtained?

826 Answer: [N/A]

827 Justification: The research involves no study participants or human-subject data.

828 **16. Declaration of LLM usage**

829 Question: Does the paper describe the usage of LLMs if it is an important, original, or  
830 non-standard component of the core methods in this research? Note that if the LLM is used  
831 only for writing, editing, or formatting purposes and does not impact the core methodology,  
832 scientific rigor, or originality of the research, declaration is not required.

833 Answer: [N/A]



834 Justification: The paper’s core method does not depend on an LLM. The main contributions—  
835 the achievability type discipline, its formal metatheory, and the deterministic checker—are  
836 defined and verified independently of LLM inference. The LLM is used only as a front-end  
837 to compact a natural-language skill into the formal pack consumed by the checker; the  
838 same formal pack can be provided directly without an LLM. The formal guarantees and the  
839 reported evaluation therefore do not depend on the behavior of any particular LLM.